1. Proof: An element of $SE(3)$ is a rigid transformation.

证明刚体变换 ①保距性 ②保向性

$g \in SE(3)$，可以表示为矩阵形式 $g = \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix}$，其中 $R \in SO(3)$ $p \in \mathbb{R}^3$

则 R 满足：$R^TR = I$, $\det(R) = 1$

对任意点 q，变换后点坐标 $g(q) = Rq + p$

对任意向量 $v = q_1 - q_2$，变换后向量为

$g_*(v) = g(q_1) - g(q_2) = Rq_1 + p - (Rq_2 + p) = R(q_1 - q_2) = Rv$

①保距性

$\|g(q_1) - g(q_2)\|^2 = \|R(q_1 - q_2)\|^2 = [R(q_1 - q_2)]^T [R(q_1 - q_2)]$

$= (q_1 - q_2)^T R^T R (q_1 - q_2) = (q_1 - q_2)^T (q_1 - q_2) = \|q_1 - q_2\|^2$

②保向性

$$g_*(v_1 \times v_2) = R(v_1 \times v_2)$$

根据 $SO(3)$ 的叉乘性质 $R(a \times b) = (Ra) \times (Rb)$

则有 $R(v_1 \times v_2) = R(v_1) \times (Rv_2) = g_*(v_1) \times g_*(v_2)$

所以 $SE(3)$ 的元素是刚体变换

2. Given a homogeneous transformation matrix and we assume that it rotates around a certain axis.

$$g = \begin{bmatrix} 0 & 1 & 0 & 10 \\ 0 & 0 & -1 & 20 \\ -1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Find the equivalent rotational axis ω, rotation angle θ, and $\xi = [v^T, \omega^T]^T$.

由 $g = \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix}$ 可提取出旋转矩阵 $R = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & -1 \\ -1 & 0 & 0 \end{bmatrix}$，移动向量 $p = \begin{bmatrix} 10 \\ 20 \\ 1 \end{bmatrix}$

旋转角度 $\theta = \arccos \frac{\text{trace}(R) - 1}{2}$

$\text{trace}(R) = 0 + 0 + 0 = 1$ $\quad \theta = \arccos \frac{-1}{2} = \frac{2\pi}{3}$

求旋转轴 ω：$\hat{\omega} = \frac{1}{2\sin\theta}(R - R^T)$

$\hat{w} = \frac{1}{2\sin\frac{\pi}{3}}\left(\begin{bmatrix}0&1&0\\0&0&1\\1&0&0\end{bmatrix} - \begin{bmatrix}0&0&1\\1&0&0\\0&1&0\end{bmatrix}\right) = \frac{1}{\sqrt{3}}\begin{bmatrix}0&1&-1\\-1&0&1\\1&-1&0\end{bmatrix}$

从 $\hat{w}$ 提取轴线 $w = \frac{1}{\sqrt{3}}\begin{bmatrix}1\\1\\-1\end{bmatrix}$

求线速度分量 v：先找出螺旋的节距 h 和轴上一点 q，

再套用 $v = -w\times q + hw$

S[3] 平移部分：$p = (I-R)q + h\theta w$

$h = \frac{w^T p}{\theta} = \frac{\frac{1}{\sqrt{3}}\begin{bmatrix}1&1&-1\end{bmatrix}\begin{bmatrix}\alpha\\0\\1\end{bmatrix}}{\frac{2\pi}{3}} = \frac{2\sqrt{3}}{2\pi}$

由公式 $(I-R)q = p - h\theta w$ 有

$\begin{bmatrix}1&-1&0\\0&1&1\\1&0&1\end{bmatrix}\begin{bmatrix}q_1\\q_2\\q_3\end{bmatrix} = \begin{bmatrix}\frac{1}{2}\\\frac{\sqrt{3}}{2}\\0\end{bmatrix}$ 一个解为 $q = \begin{bmatrix}\frac{\sqrt{3}}{3}\\0\\0\end{bmatrix}$

则 $v = -w\times q + hw = \frac{1}{\sqrt{3}}\begin{bmatrix}1\\1\\-1\end{bmatrix}\times\begin{bmatrix}\frac{\sqrt{3}}{3}\\0\\0\end{bmatrix} + \frac{2\sqrt{3}}{2\pi}\cdot\frac{1}{\sqrt{3}}\begin{bmatrix}1\\1\\-1\end{bmatrix} = \begin{bmatrix}-\frac{\sqrt{3}}{9}+\frac{2}{2\pi}\\\frac{\sqrt{3}}{9}+\frac{2}{2\pi}\\\frac{\sqrt{3}}{9}-\frac{2}{2\pi}\end{bmatrix}$

综上 $\xi = \begin{bmatrix}-\frac{\sqrt{3}}{9}+\frac{2}{2\pi}\\\frac{\sqrt{3}}{9}+\frac{2}{2\pi}\\\frac{\sqrt{3}}{9}-\frac{2}{2\pi}\\\frac{\sqrt{3}}{3}\\0\\0\end{bmatrix}$

3. The following frame definitions are given as known

$$g_{ua} = \begin{bmatrix}\sqrt{3}/2 & -1/2 & 0 & 1\\ 1/2 & \sqrt{3}/2 & 0 & -1\\ 0 & 0 & 1 & 8\\ 0 & 0 & 0 & 1\end{bmatrix}$$

$$g_{ba} = \begin{bmatrix}1 & 0 & 0 & 0\\ 0 & \sqrt{3}/2 & -1/2 & 10\\ 0 & 1/2 & \sqrt{3}/2 & -20\\ 0 & 0 & 0 & 1\end{bmatrix}$$

$$g_{cu} = \begin{bmatrix}\sqrt{3}/2 & -1/2 & 0 & -3\\ \sqrt{3}/4 & 3/4 & -1/2 & -3\\ 1/4 & \sqrt{3}/4 & \sqrt{3}/2 & 3\\ 0 & 0 & 0 & 1\end{bmatrix}$$

Solve for g_{cb}.

$$g_{cb} = g_{ca} \cdot g_{ab} = g_{ca} \cdot g_{ac} \cdot (g_{ba})^{-1}$$

$$= \begin{bmatrix} 0 & -1 & 0 & -3 \\ 0 & 0 & -1 & -7 \\ 1 & 0 & 0 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 & 1 \\ 0 & 0 & -1 & 2 \\ 1 & 0 & 0 & 8 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}^{-1}$$

$$= \begin{bmatrix} 0 & -1 & 0 & -4 \\ 0 & 0 & -1 & -5 \\ 1 & 0 & 0 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

4. Find the homogeneous transformation matrix g_{ab} of the frame $\{B\}$ with respect to the frame $\{A\}$, which is generated by the following sequence of rigid motion.
   (1) Rotation about Z_A by 90 degrees;
   (2) Rotation about X_A by -90 degrees;
   (3) Translation to the vector $(3,7,9)^T$.

相对固定坐标系的连续变换——左乘

$$R = R_x(-90) \cdot R_z(90) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(-90) & -\sin(-90) \\ 0 & \sin(-90) & \cos(-90) \end{bmatrix} \cdot \begin{bmatrix} \cos 90 & -\sin 90 & 0 \\ \sin 90 & \cos 90 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & 1 \\ -1 & 0 & 0 \end{bmatrix} \qquad p = \begin{bmatrix} 3 \\ 7 \\ 9 \end{bmatrix}$$

$$g_{ab} = \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 & 3 \\ 0 & 0 & 1 & 7 \\ -1 & 0 & 0 & 9 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

5. (a) Referring to Fig. 1, give the value of g_{ab};

(b) Referring to Fig. 1, give the value of g_{ac};

(c) Referring to Fig. 1, give the value of g_{bc};

(d) Referring to Fig. 1, give the value of g_{ca}.

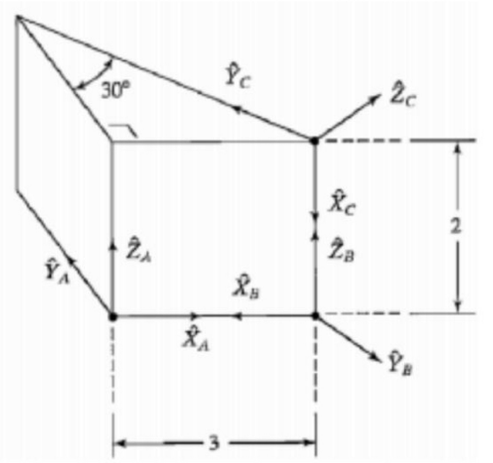


Fig. 1. Frames at the corners of a wedge

(a) g_{ab}：B相对A的位姿　在A看B

R_{ab}：用B的基向量在A下的坐标，即可拆开写出来

$$R_{ab} = \begin{bmatrix} \hat{X}_B^A & \hat{Y}_B^A & \hat{Z}_B^A \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

P_{ab}：在A看B的原点的位置

$$P_{ab} = [3 \quad 0 \quad 0]^T$$

$$g_{ab} = \begin{bmatrix} R_{ab} & P_{ab} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 & 3 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

(b) $$R_{ac} = \begin{bmatrix} 0 & -\sin 30^\circ & \cos 30^\circ \\ 0 & \cos 30^\circ & \sin 30^\circ \\ -1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ 0 & \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -1 & 0 & 0 \end{bmatrix}$$

$$P_{ac} = [3 \quad 0 \quad 2]^T$$

$$g_{ac} = \begin{bmatrix} R_{ac} & P_{ac} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -\frac{1}{2} & \frac{\sqrt{3}}{2} & 3 \\ 0 & \frac{\sqrt{3}}{2} & \frac{1}{2} & 0 \\ -1 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

(c) $$R_{bc} = \begin{bmatrix} 0 & \sin 30^\circ & -\cos 30^\circ \\ 0 & -\cos 30^\circ & -\sin 30^\circ \\ -1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ 0 & -\frac{\sqrt{3}}{2} & -\frac{1}{2} \\ -1 & 0 & 0 \end{bmatrix}$$

$$P_{bc} = [0 \quad 0 \quad 2]^T$$

$$g_{bc} = \begin{bmatrix} 0 & \frac{1}{2} & -\frac{\sqrt{3}}{2} & 0 \\ 0 & -\frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \\ -1 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

(d) $g_{ca} = g_{ac}^{-1} =$

$$\begin{bmatrix} 1 & 0 & 0 & -\frac{\sqrt{3}}{2} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$